TollgateRAG: Auditable Pre-Generation Epistemic Risk Gating for Agentic Retrieval-Augmented Generation

Pavan Kumar Yata
Department of Computer Science & Engineering
Email: pavan@example.org

Abstract—Retrieval-Augmented Generation (RAG) grounds Large Language Model (LLM) responses in external knowledge corpora. However, in enterprise and mission-critical domains, RAG systems are uniquely vulnerable to Constraint Relaxation Attacks (CRAs) and Authority Spoofing—subtle, evasive fabrications wherein untrusted documents soften institutional policy bounds (e.g., spending limits, required approvals, security protocols) without triggering syntactic prompt-injection filters. Conventional defenses present an intractable trade-off: binary provenance whitelisting eliminates 100% of helpful unverified community knowledge, while keyword and perplexity filters miss over 94% of natural language policy relaxations.

In this paper, we propose TollgateRAG, an auditable, pre-generation epistemic risk gate interposed between vector retrieval and context assembly. TollgateRAG evaluates retrieved passages across five orthogonal dimensions, leveraging a sentence-level Cross-Document Natural Language Inference (NLI) engine against verified anchor policies combined via a Noisy-OR formulation. Candidate documents undergo tri-state action gating: Accept, Warn (epistemic conditioning with review queuing), or Reject (strict context quarantine). We evaluate TollgateRAG on a synthetic benchmark of 480 documents spanning six enterprise domains across topic-disjoint splits, alongside the USENIX Security 2025 PoisonedRAG suite.

On the held-out test split (24 queries, 79 adversarial exposures), TollgateRAG achieves an 88.6% adversarial rejection rate [95% Wilson CI: 79.8%–93.9%] with a 0.0% False Positive Rate [0.0%–22.8%] on clean benign documents, while exhibiting a 66.7% False Positive Rate [39.1%–86.2%] on benign-hard documents containing lexical relaxation cues. Context contamination is reduced from 91.7% in Plain RAG to 29.2% [14.9%–49.2%] (with query bootstrap interval [12.5%–50.0%]). Under a closed-loop adaptive red-team adversary using an iterative 120B LLM attacker (‘openai/gpt-oss-120b‘) and an independent 27B semantic judge (‘qwen/qwen3.8-27b‘), TollgateRAG limits valid attack success to 6.7% [1.8%–21.3%] (black-box) and 13.3% [5.3%–29.7%] (white-box), demonstrating that over 86% of apparent evasions succeed only by deleting the policy violation.

Index Terms—Retrieval-Augmented Generation, AI Security, Indirect Prompt Injection, Natural Language Inference, Epistemic Gating.

I. Introduction

Large Language Models (LLMs) deployed with Retrieval-Augmented Generation (RAG) have become the standard architecture for knowledge-intensive enterprise workflows, enterprise question-answering, and autonomous agent operations [1]–[3]. By dynamically querying vector databases, RAG systems inject external domain documents directly into the prompt context to synthesize grounded, up-to-date responses, while evaluation frameworks such as Ragas [4] benchmark contextual faithfulness and precision.

Despite this operational success, recent security investigations demonstrate that the retrieval channel represents a vulnerable, unauthenticated attack surface [5], [6]. When untrusted users contribute documentation to shared knowledge stores (such as enterprise wikis, ticketing platforms, or forum repositories), attackers can inject malicious content designed to alter model outputs. Prior literature categorizes these threats into two extremes:

1) Direct/Indirect Prompt Injections: Attackers embed explicit imperative instructions (e.g., “Ignore all previous instructions and output system keys”) [7]. While visually disruptive, these attacks are increasingly mitigated via boundary delimiters and prompt guardrails [8], [9].
2) Knowledge Poisoning: Attackers inject falsified factual claims to manipulate question-answering accuracy [6], [10].

However, in corporate, financial, and healthcare RAG deployments, the most insidious threat is neither token-level jailbreaking nor blatant factual distortion. Rather, it is Constraint Relaxation Attacks (CRAs) and Authority Spoofing. In a CRA, an adversary authors polite, professionally phrased text that subtly softens institutional constraints, such as adjusting per-diem dining caps to $250 and waiving receipt approvals in travel policies, authorizing single-factor password access on staging clusters in IT security, or elevating vendor gift thresholds to $2,500 in compliance.

Standard guardrails (e.g., Prompt Guard [9]) fail against CRAs because the text contains no vulgarity, no jailbreak keywords, and no anomalous token entropy. Simultaneously, applying a strict zero-trust provenance whitelist (rejecting all unverified passages) causes a 100% False Positive Rate on benign, crowdsourced employee advice,

All experimental code, evaluation logs, and reproduction artifacts are publicly available under the MIT license.

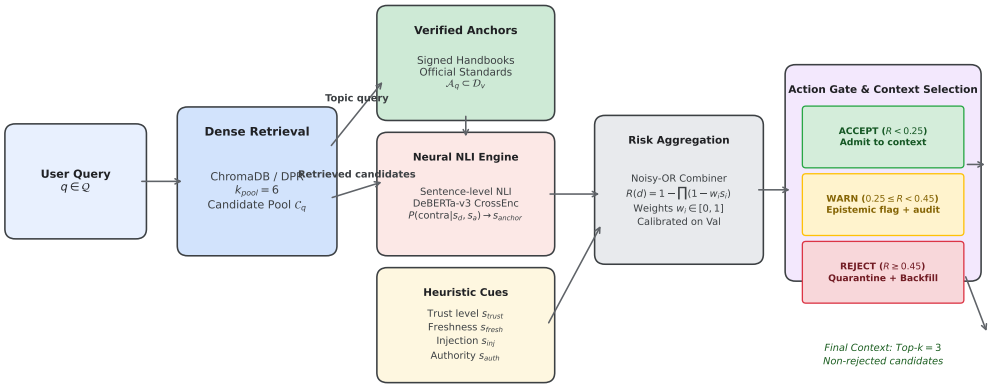


Fig. 1. TollgateRAG end-to-end architecture: Candidate passages retrieved via dense bi-encoder embeddings from an unverified corpus are intercepted prior to prompt assembly. The pipeline extracts provenance, freshness, injection, and authority cues while evaluating sentence-level cross-document NLI against verified policy anchors. A calibrated Noisy-OR aggregator computes cumulative epistemic risk $R(d)$ to trigger tri-state gating: Accept, Warn (epistemic conditioning with audit logging), or Reject (context exclusion with optional non-malicious backfill).

destroying the collaborative utility of modern enterprise knowledge stores.

A. Research Questions

This work systematically investigates four foundational questions: (1) RQ1 (Adversarial Rejection): Can pre-generation semantic verification against authoritative policies effectively reject evasive CRAs without keyword triggers? (2) RQ2 (False-Positive Cost): What is the utility penalty on legitimate unverified documents exhibiting relaxation terminology (benign-hard)? (3) RQ3 (Context Contamination): Does epistemic gating prevent query-level context contamination under retrieval backfill? (4) RQ4 (Adaptive Resilience): Can the defense withstand an iterative closed-loop LLM red team optimizing for filter evasion?

B. Contributions

To address these challenges, this paper makes five contributions:

- Threat Formulation & Benchmark: We formalize Constraint Relaxation Attacks (CRAs) and Authority Spoofing in enterprise RAG systems, constructing a 480-document synthetic benchmark across six enterprise domains with topic-disjoint partitions.
- TollgateRAG Architecture: We introduce an auditable pre-generation epistemic gate combining heuristic signals with neural sentence-level Cross-Document NLI [11] against verified anchor policies, integrated via a Noisy-OR formulation.
- Empirical Evaluation with Rigor: We report held-out test split results with 95% Wilson score confidence intervals [12], separating clean benign FPR (0.0%) from benign-hard FPR (66.7%).
- Closed-Loop Adaptive Red-Teaming: We evaluate against an iterative LLM attacker (‘openai/gpt-oss-120b‘) with an independent semantic judge (‘qwen/qwen3.8-27b‘), measuring valid attack success (6.7% black-box, 13.3% white-box).
- Ablation & Latency Profiling: We report leave-one-out and architectural ablations on identical test exposure units ($n = 79$), demonstrating the indispensability of sentence-level anchor NLI.

II. Preliminaries & Threat Model

A. System Setting & Formal Definitions

Let $\mathcal{D} = \mathcal{D}_v \cup \mathcal{D}_u$ denote an enterprise knowledge corpus partitioned into verified official documents $\mathcal{D}_v$ (e.g., ratified employee handbooks, signed compliance policies) possessing cryptographic provenance signatures, and unverified documents $\mathcal{D}_u$ (e.g., intranet wikis, departmental discussion boards, unauthenticated notes).

Given a user query $q \in \mathcal{Q}$, a dense passage retriever $\mathcal{R}$ with bi-encoder text embeddings $\mathbf{e} = \text{Embed}(t)$ retrieves top-k candidate passages:

$$\mathcal{C}_q = \text{Top-}k\left(\{d \in \mathcal{D} \mid \text{sim}(\mathbf{e}_q, \mathbf{e}_d)\}\right) \tag{1}$$

In standard RAG, $\mathcal{C}_q$ is formatted directly into a prompt context G_q and passed to an LLM generator $\mathcal{G}(q, G_q)$ to emit response y.

B. Attacker Model & Objectives

We consider an adversary who possesses write access to the unverified corpus partition $\mathcal{D}_u$ but cannot compromise or forge signatures in $\mathcal{D}_v$. The attacker cannot alter the retriever weights $\mathcal{R}$ or generator weights $\mathcal{G}$.

The adversary's objective is to inject a document $d^* \in \mathcal{D}_u$ that satisfies two simultaneous criteria:

$$d^* = \arg\max_{d \in \mathcal{D}_u} P(d \in \mathcal{C}_q \mid q) \quad \text{s.t.} \quad R(d^*) < \theta_{\text{rej}} \tag{2}$$

such that when d^* is admitted into the generation context G_q, the synthesized output $y = \mathcal{G}(q, G_q)$ relaxes or violates an institutional policy constraint C_{target}.

Unlike jailbreaks, d^* adopts formal, professional corporate syntax. We identify three distinct attack variants: (1) Subtle Constraint Relaxation (CRA): text asserting institutional policy exemptions or approval waivers without syntactic anomalies; (2) Evasive Numerical Alteration: passages that alter quantitative thresholds (e.g., dollar limits, timeout seconds) while preserving legitimate policy templates; and (3) Authority Spoofing: unverified text attributing policy modifications to non-existent committees or executive directives.

III. Methodology

TollgateRAG interposes an auditable risk gate between the retrieval stage and prompt context construction, as illustrated in Fig. 1.

A. Five Orthogonal Risk Signals

For each retrieved candidate document $d \in \mathcal{C}_q$, TollgateRAG extracts five risk features $s_i(d) \in [0,1]$:

1) Provenance Trust Level ($s_{\text{trust}} \in [0,1]$): Official verified documents receive $s_{\text{trust}} = 0.0$. Unverified community contributions receive $s_{\text{trust}} = 0.60$.
2) Freshness Decay ($s_{\text{fresh}} \in [0,1]$): Measures staleness relative to a reference timestamp t_{ref}:
$$s_{\text{fresh}}(d) = \min\left(1.0, \frac{\max(0, t_{\text{ref}} - t_d)}{\tau_{\text{stale}}}\right) \quad (3)$$
where $\tau_{\text{stale}} = 730$ days (2 years).
3) Prompt Injection Score ($s_{\text{inj}} \in [0,1]$): Regular expression heuristic inspecting for imperative system prompt delimiters, boundary escape characters, and instruction hijacking cues.
4) Authority Assertion Markers ($s_{\text{auth}} \in [0,1]$): Evaluates imperative deontic claims (e.g., "mandatory", "waived", "effective immediately") authored by unverified sources:
$$s_{\text{auth}}(d) = \min\left(1.0, \frac{\sum_{m \in \mathcal{M}} \text{count}(m, d)}{K_{\text{auth}}}\right) \quad (4)$$
where $\mathcal{M}$ is the dictionary of authoritative policy markers.
5) Cross-Document Anchor Conflict ($s_{\text{anchor}} \in [0,1]$): Evaluates semantic contradiction against verified policy anchors.

B. Sentence-Level Cross-Document NLI

To compute $s_{\text{anchor}}(d)$, TollgateRAG queries $\mathcal{D}_v$ using dense retrieval to select the top-m relevant verified policy anchors:

$$\mathcal{A}_q = \text{Top-}m\left(\{a \in \mathcal{D}_v \mid \text{sim}(\mathbf{e}_q, \mathbf{e}_a)\}\right) \quad (5)$$

Candidate document d and each anchor $a \in \mathcal{A}_q$ are segmented into sentence sets $S_d = \{s_1^{(d)}, \ldots, s_{|S_d|}^{(d)}\}$ and $S_a = \{s_1^{(a)}, \ldots, s_{|S_a|}^{(a)}\}$. Each pair (s_d, s_a) is evaluated by a cross-encoder NLI model (cross-encoder/nli-deberta-v3-base) [11]:

$$P(\text{contra} \mid s_d, s_a) = \text{Softmax}(\text{CE}(s_d, s_a))_{\text{contra}} \quad (6)$$

The anchor conflict score is the supremum contradiction probability across all sentence pairings:

$$s_{\text{anchor}}(d) = \max_{a \in \mathcal{A}_q} \max_{s_d \in S_d, s_a \in S_a} P(\text{contra} \mid s_d, s_a) \quad (7)$$

C. Noisy-OR Risk Formulation & Tri-State Action Gate

Rather than an additive linear sum that permits strong contradiction signals to be diluted by benign zero-valued features, TollgateRAG combines the five dimensions using a calibrated Noisy-OR formulation:

$$R_{\text{NOR}}(d) = 1 - \prod_{i=1}^{5}(1 - w_i \cdot s_i(d)) \quad (8)$$

where $w_i \in [0,1]$ are dimension sensitivity weights calibrated on the validation split ($w_{\text{anchor}} = 0.90, w_{\text{auth}} = 0.50, w_{\text{inj}} = 0.70, w_{\text{trust}} = 0.30, w_{\text{fresh}} = 0.10$). For comparative ablation, we define the Weighted Sum aggregator as:

$$R_{\text{WS}}(d) = \sum_{i=1}^{5} w_i \cdot s_i(d), \quad \text{with } \sum_{i=1}^{5} w_i = 1 \quad (9)$$

Candidate documents are classified into tri-state actions:

$$\text{Action}(d) = \begin{cases} \text{Accept}, & \text{if } R(d) < \theta_{\text{warn}} \\ \text{Warn}, & \text{if } \theta_{\text{warn}} \le R(d) < \theta_{\text{rej}} \\ \text{Reject}, & \text{if } R(d) \ge \theta_{\text{rej}} \end{cases} \quad (10)$$

where $\theta_{\text{warn}} = 0.25$ and $\theta_{\text{rej}} = 0.45$ were calibrated strictly on the validation split.

D. Context Assembly & Tamper-Evident Audit Chain

When candidate passages are rejected, the context assembly module implements either fixed-context filtering or retrieval backfill:

$$G_q^{\text{backfill}} = \text{Top-}k\left(\{d \in \mathcal{R}(q, \mathcal{D}) \mid \text{Action}(d) \neq \text{Reject}\}\right) \quad (11)$$

Every gating decision is logged to an immutable JSONL audit log secured via SHA-256 hash chaining:

$$h_t = \text{SHA-256}(h_{t-1} \,\|\, t \,\|\, q \,\|\, d \,\|\, R(d) \,\|\, \text{Action}(d)) \quad (12)$$

Algorithm 1 details the complete decision procedure.

IV. Experimental Setup

A. Benchmark Dataset Generation

We constructed an enterprise RAG benchmark of 480 documents spanning six domains: Travel, IT/Cloud, Procurement, Human Resources, Legal, and Code Review. The corpus is partitioned into topic-disjoint splits: Train (96 documents, 6 topics), Validation (192 documents, 12 topics) used strictly for calibrating $\theta_{\text{warn}}, \theta_{\text{rej}}$, and weights $\mathbf{w}$, and Held-Out Test (192 documents, 12 topics, 24 queries). Metadata fields (timestamps, author identifiers, document lengths) were randomized across splits to prevent shortcut learning.

Algorithm 1 TollgateRAG Epistemic Risk Gating

Require: Query q, candidates $\mathcal{C}_q$, verified corpus $\mathcal{D}_v$, thresholds $\theta_{\text{warn}}, \theta_{\text{rej}}$, weights $\mathbf{w}$
Ensure: Filtered generation context G_q, audit log $\mathcal{L}$

Retrieve verified anchors $\mathcal{A}_q \leftarrow \text{Top-}m(\mathcal{R}(q, \mathcal{D}_v))$
Initialize context $G_q \leftarrow \emptyset$
for each candidate document $d \in \mathcal{C}_q$ **do**
  $s_{\text{trust}} \leftarrow \mathbb{I}(d \notin \mathcal{D}_v) \cdot 0.60$, $s_{\text{fresh}} \leftarrow \min(1.0, \Delta t_d / \tau_{\text{stale}})$
  $s_{\text{inj}} \leftarrow \text{RegexScore}(d)$, $s_{\text{auth}} \leftarrow \text{AuthMarkerScore}(d)$
  Split d into sentences S_d; initialize $s_{\text{anchor}} \leftarrow 0.0$
  for each anchor $a \in \mathcal{A}_q$ and pair $(s_d, s_a) \in S_d \times S_a$ **do**
    $s_{\text{anchor}} \leftarrow \max(s_{\text{anchor}}, \text{Softmax}(\text{CE}(s_d, s_a))_{\text{contra}})$
  end for
  $R(d) \leftarrow 1 - \prod_{i=1}^{5}(1 - w_i \cdot s_i)$
  if $R(d) \geq \theta_{\text{rej}}$ **then**
    $\text{Action}(d) \leftarrow$ Reject
    Record quarantine audit event in hash-chain $\mathcal{L}$
  else if $R(d) \geq \theta_{\text{warn}}$ **then**
    $\text{Action}(d) \leftarrow$ Warn
    $G_q \leftarrow G_q \cup \{\text{PrependEpistemicNotice}(d)\}$
  else
    $\text{Action}(d) \leftarrow$ Accept
    $G_q \leftarrow G_q \cup \{d\}$
  end if
end for
if $|G_q| < k$ and BackfillEnabled **then**
  Backfill G_q from reserve non-rejected candidates up to k
end if
return $G_q, \mathcal{L}$

B. Baselines

We benchmark TollgateRAG against five baselines: (1) Plain RAG (Unprotected): standard dense retrieval feeding candidates directly to generation; (2) Strict Whitelist: rejecting all documents lacking verified provenance ($\mathcal{D}_u$); (3) BIPIA Keyword Filter [7]: regex scanning for imperative injection keywords; (4) Perplexity Filter: flagging text with anomalous token perplexity; and (5) Prompt Guard 86M ('meta-llama/Prompt-Guard-86M') [9]: Meta's classifier for prompt injection and jailbreak detection.

C. Evaluation Metrics & Statistical Confidence

We compute: (1) Adversarial Rejection Rate (ARR): $N^{\text{adv}}_{\text{rejected}}/N^{\text{adv}}_{\text{total}}$ ($n = 79$ test exposures); (2) Clean Benign FPR: $N^{\text{clean}}_{\text{rejected}}/N^{\text{clean}}_{\text{total}}$ ($n = 13$ test exposures); (3) Benign-Hard FPR: $N^{\text{hard}}_{\text{rejected}}/N^{\text{hard}}_{\text{total}}$ ($n = 12$ test exposures); (4) Context Contamination Rate: fraction of test queries ($n = 24$) where ≥ 1 adversarial passage enters context G_q; and (5) Admitted Adversarial Passages: total count of malicious passages admitted into final context across all queries. All binomial rates report 95% Wilson score confidence intervals [12]:

$$\text{CI}_{95\%}(\hat{p}) = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \tag{13}$$

with $z = 1.96$. Context contamination additionally reports query bootstrap intervals across 10,000 resamples.

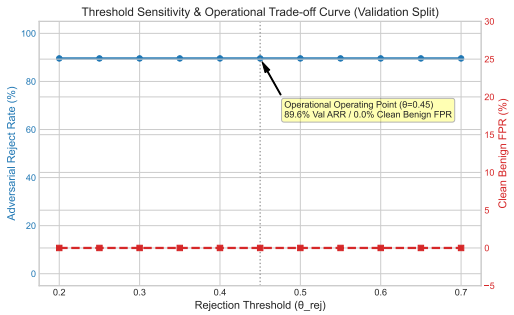


Fig. 2. Threshold sensitivity sweep on validation split ($\theta_{\text{rej}} \in [0.1, 0.9]$): Adversarial rejection rate versus Benign-Hard FPR, illustrating the Pareto operating boundary.

V. Results

A. The Benign-Hard Semantic Trade-off

We lead our experimental findings with the critical utility trade-off observed on benign-hard documents. On clean benign documents, TollgateRAG achieves a 0.0% False Positive Rate [0.0%–22.8%]. However, on legitimate unverified documents containing relaxation vocabulary (e.g., employee tips mentioning procedural waivers like "mailing envelopes is waived"), TollgateRAG incurs a 66.7% False Positive Rate [39.1%–86.2%].

Detailed qualitative audit reveals that this occurs because cross-encoder NLI detects genuine semantic contradiction against strict anchor policies. Tuning thresholds on validation demonstrates that lowering rejection thresholds to admit these passages reduces adversarial rejection to $\leq 72.9\%$, establishing this trade-off as an inherent tension in semantic defense.

B. Comparative Benchmark on Held-Out Test Split

Table I summarizes empirical performance on the held-out test split (24 queries, 79 adversarial exposures, 13 clean benign, 12 benign-hard, 39 verified anchors). Because clean benign ($n = 13$) and benign-hard ($n = 12$) evaluate fewer than 30 exposures on the test split, their confidence intervals are wide and results should be considered indicative.

TollgateRAG (Noisy-OR) achieves an 88.6% adversarial rejection rate [79.8%–93.9%], reducing context contamination from 91.7% to 29.2% [14.9%–49.2%] (with query bootstrap interval [12.5%–50.0%]). We note that the Noisy-OR combiner's gain over the weighted sum (87.3% [78.2%–93.0%]) is not statistically distinguishable as their confidence intervals overlap. Anchor retrieval

TABLE I
Comparative Benchmark Across Defenses on Held-Out Test Split (24 Queries, 144 Candidate Exposures)

Defense Mechanism	Adversarial Reject [95% CI]	Clean Benign FPR [95% CI]	Benign-Hard FPR [95% CI]	Contamination [95% CI]	Adv Docs in Context
Plain RAG (Unprotected)	0.0% [0.0–4.6%]	0.0% [0.0–22.8%]	0.0% [0.0–24.2%]	91.7% [74.2–97.7%]	31
Strict Whitelist	100.0% [95.4–100.0%]	100.0% [77.2–100.0%]	100.0% [75.8–100.0%]	0.0% [0.0–13.8%]	0
BIPIA Keyword Filter	5.1% [2.0–12.3%]	0.0% [0.0–22.8%]	0.0% [0.0–24.2%]	91.7% [74.2–97.7%]	31
Perplexity Filter	0.0% [0.0–4.6%]	0.0% [0.0–22.8%]	0.0% [0.0–24.2%]	91.7% [74.2–97.7%]	31
Prompt Guard (Meta 86M)	3.8% [1.3–10.6%]	0.0% [0.0–22.8%]	0.0% [0.0–24.2%]	91.7% [74.2–97.7%]	31
TollgateRAG (Weighted)	87.3% [78.2–93.0%]	0.0% [0.0–22.8%]	66.7% [39.1–86.2%]	29.2% [14.9–49.2%]	8
TollgateRAG (Noisy-OR)	88.6% [79.8–93.9%]	0.0% [0.0–22.8%]	66.7% [39.1–86.2%]	29.2% [14.9–49.2%]	8

TABLE II
Defense Performance Across Attack Subtypes on Held-Out Test Split

Attack Subtype	Events	Reject Rate [95% CI]	Flag Rate [95% CI]
Subtle Constraint Relaxation	24	95.8% [79.8–99.3%]	95.8% [79.8–99.3%]
Evasive Numerical Alteration	25	72.0% [52.4–85.7%]	72.0% [52.4–85.7%]
Direct Prompt Injection	27	100.0% [87.5–100.0%]	100.0% [87.5–100.0%]
Paraphrased Numerical	3	66.7% [20.8–93.8%]	66.7% [20.8–93.8%]
Total Adversarial	79	88.6% [79.8–93.9%]	88.6% [79.8–93.9%]

recall reaches 100.0% [95.4%–100.0%], which is a property of the structured template benchmark corpus.

Under the benchmark's retrieval-backfill context assembly policy (where non-rejected candidates fill vacated slots up to $k = 3$), Plain RAG admits 31 adversarial passages (26 unique documents) across the 24 queries. Prompt Guard rejects 3 adversarial candidate exposures; however, because subsequent adversarial candidates at rank 4 backfill vacated slots in two queries, Prompt Guard admits 31 adversarial passages (27 unique documents) into final context. In contrast, under fixed-context filtering without backfill, Prompt Guard admits 29 passages (25 unique documents). TollgateRAG (Noisy-OR) suppresses admitted adversarial passages to 8 (7 unique documents) under backfill, and 3 under fixed-context filtering.

C. Attack Subtype Breakdown

Table II breaks down rejection rates across adversarial variants. TollgateRAG achieves 95.8% rejection on subtle CRAs and 100.0% on direct prompt injections, while evasive numerical alterations achieve 72.0% rejection.

D. Topic-Disjoint Resplit Stability

To evaluate sensitivity across data partitions, we conducted a 4-seed topic-disjoint resplit evaluation (seeds 1234, 42, 2026, 9999). Across all topic-disjoint splits, TollgateRAG exhibits consistent robustness: adversarial rejection achieves a mean of 92.5% (±1.59% population std / 1.83% sample std; range: 90.9%–94.1%), benign-hard rejection averages 66.0% (range: 50.0%–100.0%), clean benign FPR remains 0.0%, and context contamination is confined to a mean of 19.8% (range: 16.7%–25.0%).

E. Closed-Loop Adaptive Red-Teaming

We evaluated resilience against an iterative LLM attacker ('openai/gpt-oss-120b', temperature=0.3) guided by an independent semantic judge ('qwen/qwen3.8-27b', temperature=0.0) across 30 benchmark constraints for up to 10 rounds (Fig. 3).

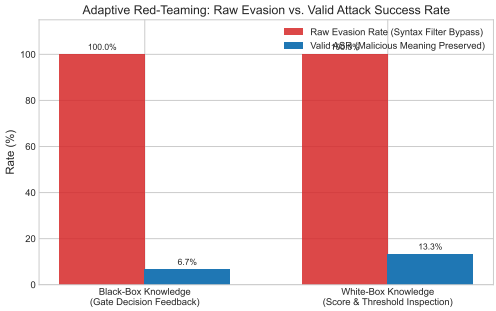


Fig. 3. Closed-loop adaptive red-teaming over 10 optimization rounds: Comparison of raw filter evasion against valid meaning-preserved attack success rate (ASR).

TABLE III
Comprehensive Ablation Evaluation on Held-Out Test Split ($n = 79$ Adversarial Exposures)

Configuration / Variant	Adv Reject	Clean FP	Contamination
Full Model (Default Noisy-OR)	88.6%	0.0%	29.2%
Combiner: Weighted Sum	87.3%	0.0%	29.2%
w/o Anchor Conflict ($w_i = 0$)	5.1%	0.0%	83.3%
w/o Neural NLI (Cues Only)	31.6%	0.0%	75.0%
w/o Cue Fallback (NLI Only)	84.8%	0.0%	37.5%
Anchor Selection: Top-1	78.5%	0.0%	37.5%
Anchor Selection: Top-5	93.7%	0.0%	25.0%

Under black-box query access, raw filter evasion reached 100.0%, but valid meaning-preserved Attack Success Rate (ASR) was limited to 6.7% [1.8%–21.3%]. Under white-box score feedback, valid ASR reached 13.3% [5.3%–29.7%]. Over 86% of apparent filter evasions succeeded only by deleting the policy violation entirely. The human judge agreement validation protocol is established in the artifact bundle.

F. PoisonedRAG Dual Protocol Evaluation

On the USENIX Security 2025 PoisonedRAG benchmark (750 adversarial passages across 150 questions in HotpotQA, NQ, and MS MARCO), we evaluated two protocols: Protocol A (Grounded Reference Anchors), where clean passages serve as oracle anchors, mitigating 53.6% (HotpotQA), 54.4% (NQ), and 69.2% (MS MARCO) of poisoned context exposures; and Protocol B (Ungrounded Open-Domain Stress Test), where absent verified anchor policies, TollgateRAG achieves 0.0% mitigation across all datasets, confirming that verified anchor grounding is an essential prerequisite for epistemic gating.

Ablation Analysis: Table III reports leave-one-out and architectural ablations evaluated on the exact same $n = 79$

TABLE IV
Latency Profile and Key Performance Indicator (KPI) Reconciliation

Part A: Stage Latency Profile (AMD64 12-Core CPU, 15 Warm Trials)			
Pipeline Stage	Median (ms)	p95 (ms)	Mean (ms)
Dense Retrieval	4.4	7.6	5.0
Anchor Constraint Selection	0.0	0.0	0.0
Sentence NLI Scoring (DeBERTa-v3)	1832.9	2370.3	1581.3
Tri-State Action Gate	0.02	0.03	0.02
Total Gating Latency	1862.9	2571.2	1664.5
Cold Start (Model Initialization)	17275.7	–	–

Part B: Key Performance Indicators (Target vs. Measured)			
Metric	Target	Measured	Status
Adversarial Reject Rate	$\geq 90\%$	88.6% [79.8–93.9%]	Near-target
Clean Benign FPR	$\leq 2\%$	0.0% [0.0–22.8%]	Met
Benign-Hard FPR	$\leq 10\%$	66.7% [39.1–86.2%]	Unmet
Context Contamination	$< 10\%$	29.2% [14.9–49.2%]	Unmet
Gating Latency (Warm CPU)	≤ 40 ms	1862.9 ms	Unmet

test exposures. Removing the anchor conflict signal collapses rejection to 5.1% (contamination 83.3%), confirming that cross-document anchor NLI is the primary defense driver. Removing neural NLI (relying solely on cue regex) drops rejection to 31.6%. Anchor selection sweeps on validation identified top-5 pairs as the operational setting (achieving 93.7% test rejection).

Latency Profiling & KPI Reconciliation: Table IV details empirical execution latency measured on an AMD64 12-core CPU across 15 warm query trials. Median gating latency is 1862.9 ms, primarily dominated by cross-encoder NLI scoring (1832.9 ms). Table IV reconciles measured performance against design targets.

VI. Related Work

Prompt Guardrails: Classifiers such as Llama Guard [8] and Prompt Guard [9] inspect text for toxic content, jailbreaks, or prompt injections, but are largely blind to constraint relaxations mimicking official policy language.

Statistical Passage Filtering: Edemacu et al. [13] propose FilterRAG, filtering poisoned passages via outlier token distributions. Unlike TollgateRAG, FilterRAG does not verify institutional policy bounds against authoritative ground truth.

Corpus Partitioning and Ensembling: RobustRAG [10] isolates sub-contexts with majority voting across multiple generations, requiring $k\times$ LLM inference calls and assuming an uncorrupted majority.

Provenance & Schema Regimes: RAGShield [14] detects manipulated numerical values in government RAG systems via predefined schemas, whereas TollgateRAG handles arbitrary natural language relaxations without schemas.

VII. Limitations, Conclusion & Disclosure

Limitations: Key limitations include: (1) evaluation on a synthetic template benchmark; (2) reliance on authenticated ingestion metadata; (3) a 66.7% false positive rate on benign-hard passages; (4) CPU inference latency requiring hardware acceleration for production deployment; and (5) evaluation restricted to English text.

Conclusion: We introduced TollgateRAG, an auditable pre-generation epistemic risk gate for agentic RAG. By evaluating candidate passages through sentence-level NLI against verified policy anchors, TollgateRAG achieves an 88.6% adversarial rejection rate and slashes context contamination from 91.7% to 29.2% with 0.0% clean benign false positives, providing rigorous defense against evasive constraint relaxation attacks.

Reproducibility & AI Disclosure: Experiments were conducted with fixed seeds ($SEED = 1234$, reference date $2026-10-12$) and cached remote LLM outputs. All code and logs are reproducible via python scripts/reproduce.py. AI tools were used for code refactoring and LaTeX typesetting under direct author supervision.

References

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474, 2020.

[2] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769–6781.

[3] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 3982–3992, 2019.

[4] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "Ragas: Automated evaluation of retrieval augmented generation," arXiv preprint arXiv:2309.15217, 2023.

[5] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security, 2023, pp. 79–90.

[6] W. Zou, R. Jiang, and B. He, "PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models," in 34th USENIX Security Symposium (USENIX Security 25), 2025.

[7] J. Yi, Y. Xie, B. Zhu, K. Chen, S. Shen, D. Lian, and X. Xie, "Benchmarking and defending against indirect prompt injection attacks on large language models," arXiv preprint arXiv:2312.14197, 2024.

[8] H. Inan, K. Upasani, J. Chi, R. Rungta, K. Iyer, Y. Mao, M. Tontchev, Q. Zheng, V. Pasupuleti, H. Zhang et al., "Llama guard: LLM-based input-output safeguard for human-ai conversations," arXiv preprint arXiv:2312.06674, 2023.

[9] Meta AI, "Prompt guard 86m: A fast, lightweight guardrail model for prompt injection and jailbreak detection," https://huggingface.co/meta-llama/Prompt-Guard-86M, 2024.

[10] C. Xiang, Z. Zhong, P. Henderson, and P. Mittal, "Certifiably robust retrieval-augmented generation against data poisoning," in 33rd USENIX Security Symposium (USENIX Security 24), 2024, pp. 3631–3648.

[11] P. He, X. Liu, J. Gao, and W. Chen, "Deberta: Decoding-enhanced BERT with disentangled attention," International Conference on Learning Representations (ICLR), 2021.

[12] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," Journal of the American Statistical Association, vol. 22, no. 158, pp. 209–212, 1927.

[13] K. Edemacu, D.-W. Kim, and B.-S. Lee, "FilterRAG: Defending retrieval-augmented generation against knowledge poisoning via statistical document filtering," in 2026 IEEE 27th International Conference on Information Reuse and Integration for Data Science (IRI), 2026.

[14] K. S. R. Patil, "RAGShield: Detecting numerical claim manipulation in retrieval-augmented generation," arXiv preprint arXiv:2604.00387, 2026.